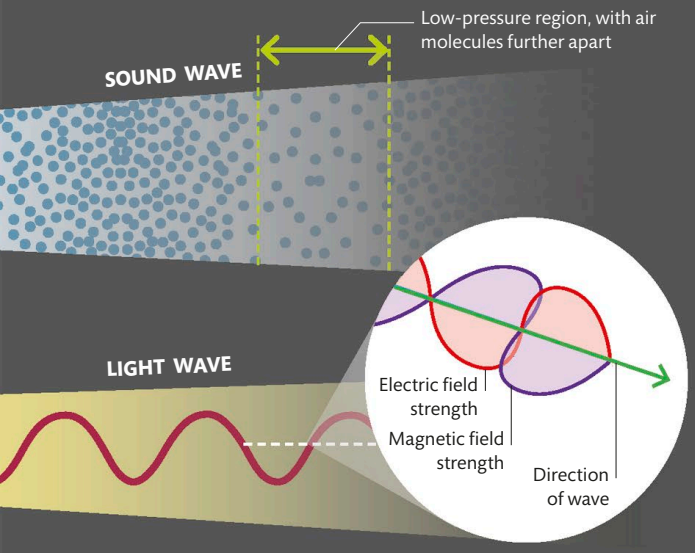
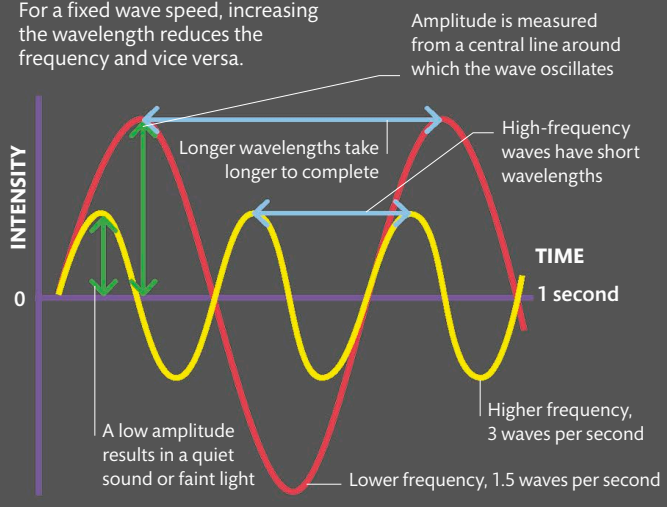
Transverse wave
Light waves are transverse: the variations in electric and magnetic fields are up and down and side to side, both at right angles to the wave's direction of travel.

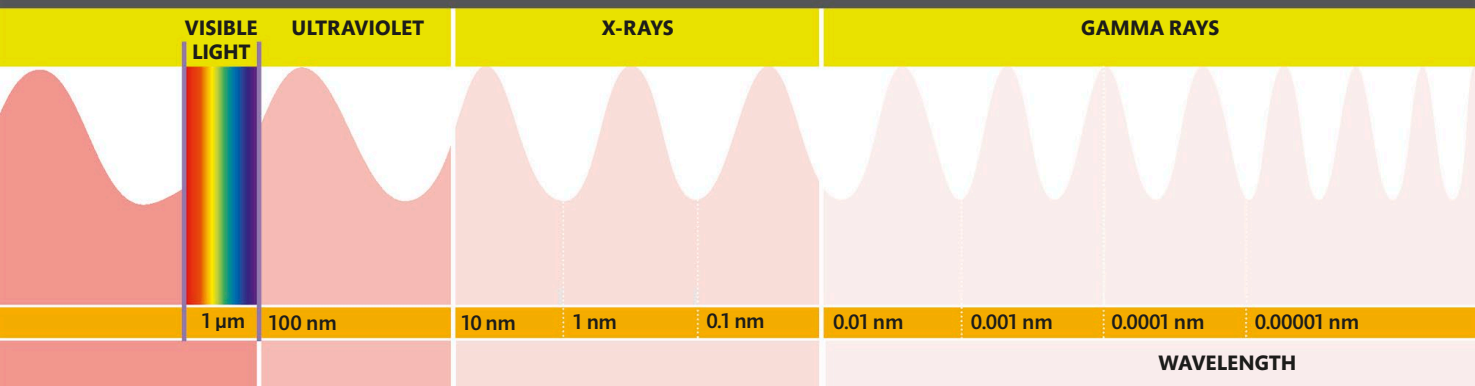


Measuring waves

All waves share measurable characteristics: speed at which they propagate (travel); amplitude (maximum intensity); frequency (how often the disturbance repeats); and wavelength (distance between waves).

Wave relationship
For a fixed wave speed, increasing the wavelength reduces the frequency and vice versa.





Human eye
Our eyes detect a narrow range of wavelengths as the color spectrum.

Disinfection
Certain wavelengths of UV light can be used to kill bacteria and sanitize objects.

Dental X-ray
Short-wavelength X-rays pass through gum tissue to reveal the teeth underneath.

Vehicle inspection
High-energy gamma rays can penetrate vehicles to reveal images of dangerous items inside.

Using electromagnetic radiation
Humans put electromagnetic radiation to use in a range of technologies. The shortest wavelengths are measured in units such as micrometers (millionths of a meter, µm) and nanometers (billionths of a meter, nm).